

Project Executable Files

Date	03 July 2026
Team ID	xxxxxx
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	3 Marks

Project Executable Files

This section requires submission of all executable and deployable files for your project. Ensure all files are properly organized, named, and uploaded to the specified submission platform. The evaluator must be able to run or deploy your solution using the provided files and instructions.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	NA
5	Environment configuration file (.env.example or similar)	NA
6	Deployed application URL (if hosted)	No
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA
8	Dockerfile / Containerization config (if applicable)	NA
9	Test files and test results	Yes

10	Demo video or walkthrough (if required)	Yes
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Step 2: File / Folder Structure

Provide an overview of your project's file and folder structure below:

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OptiCrop/
/data
  /Crop_recommendation.csv
/models
  /Crop_model.pkl
  /Kmeans_model.pkl
  /Label_encoder.pkl
  /model_name.txt
/notebooks
  /OptiCrop.ipynb
/static
  /images
    /correlation_heatmap.png
    /crop_distribution.png
    /temperature_rainfall.png
/templates
  /Index.html
  /Result.html
.gitignore
/app.py
/README.md
/Requirements.txt
/train_model.py
  
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	Not Hosted (Local Deployment)
Login Credentials (Demo)	Not Required
Platform / Hosting Provider	Local Flask Server

Repository Link	https://github.com/Harshagv123/OptiCrop
Demo Video Link	https://drive.google.com/file/d/1IntUFUXv145z3wPDsF6pHsRayfOxaNdx/view?usp=drive_link

Step 4: Run Instructions

Provide clear, step-by-step instructions for the evaluator to run your project locally:

1. Download or clone the project repository.
2. Open the project folder in Command Prompt or VS Code.
3. Create and activate a virtual environment.
4. `python -m venv myenv`
5. `myenv\Scripts\activate`
6. `pip install -r requirements.txt`
7. `python app.py`
8. <http://127.0.0.1:5000>
9. Enter soil and environmental parameters and click the prediction button to receive crop recommendations.

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Application currently runs only on local server	Can be deployed to cloud platforms in future
2	Real-time weather data integration not available	Users manually enter environmental values
3	Fertilizer recommendation feature not implemented	Planned for future enhancement